

An AI Tutor for Personalized Student Learning

Recommendations

PEAS Framework

Introduction

In this assignment, I am going to apply the PEAS framework to an AI Tutor system. PEAS stands for Performance Measure, Environment, Actuators, and Sensors. This framework is used in AI to describe how an intelligent agent works and what it needs to function properly. The AI Tutor I am analyzing is a system that looks at how a student is learning and then suggests the right content, exercises, or topics to help that student improve.

P — Performance Measure

Performance measure means how we decide if the AI Tutor is actually working well or not. In simple words, it is the goal the agent is trying to achieve.

- The student's quiz and test scores should go up over time.
- The system should be able to identify topics where the student is weak.
- The recommended lessons should actually match what the student needs.
- Students should be more engaged and spend more time studying.
- The student should feel the suggestions are helpful, not random.

E — Environment

The environment is the place or situation where the AI Tutor operates. It includes everything the agent has to deal with.

- An online learning platform like Moodle, Google Classroom, or any LMS.
- Student data such as past scores, completed lessons, and learning history.
- A collection of study material — videos, notes, quizzes, and assignments.
- Many students using the system at the same time.
- The environment keeps changing as students progress, so it is dynamic.
- The agent cannot know everything about the student, so it is partially observable.

A — Actuators

Actuators are how the AI Tutor actually does something. These are the actions it takes after analyzing the student's situation.

- It recommends the next topic or lesson the student should study.
- It generates practice questions based on where the student is struggling.

- It shows a progress report so the student can see how they are doing.
- It sends reminders or notifications to complete pending work.
- It gives feedback when a student gets an answer wrong, explaining the mistake.

S — Sensors

Sensors are how the AI Tutor collects information. Just like our eyes and ears help us understand the world, sensors help the agent understand the student.

- Quiz answers — it records which questions the student got right or wrong.
- Time spent on each lesson — it tracks where the student is spending more time.
- Login activity — it checks how often the student is using the platform.
- Student ratings or feedback about whether a lesson was useful.
- Learning preferences entered by the student, like preferred content type.
- Previous assignment scores and overall academic history.

Summary

Component	What it means	Example in AI Tutor
Performance Measure	How success is judged	Better scores, more engagement
Environment	Where the agent works	LMS, student data, content library
Actuators	What the agent does	Recommendations, quizzes, reminders
Sensors	How it collects information	Quiz results, time logs, feedback

Conclusion

After applying the PEAS framework to the AI Tutor, I can see how each component plays an important role. The performance measures keep the agent focused on helping students improve. The environment shows all the factors the agent has to work with. The actuators let it take real actions like suggesting lessons or sending reminders. And the sensors make sure it is always collecting the right data to make good decisions. Overall, the PEAS framework makes it easy to understand how this AI system is designed and what it is trying to do.